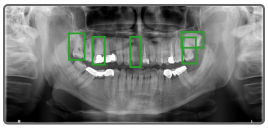
Data layer

2 datasets in use: DENTEX · Tufts
(8 more catalogued as candidates - pending)

loaders: DENTEX full · Tufts partial (1 open question)
FDI notation conversion at ingest



panoramic radiograph + per-tooth boxes + diagnoses

multi-finding ground truth
1-7 findings per image
full set - never truncated

grounding model: YOLOv8m · 5-fold CV · mAP50 ~ 0.87

1

Trace synthesis (offline)

family A generates with-tools traces
+ paired no-tools traces

tiered perturbation of displayed boxes
(nudge behavior stays demonstrable)

family B verifies + repairs every trace
staged · cross-provider · multi-finding

dual-gate decontamination
regex sweep + re-verification

corpus: 1,847 + 1,847 · 3,694 traces
exact per-cohort parity

details in the trace-synthesis figure
(all quotes, boxes and stats are real)

Training

2

1 · SFT on verified traces
behavior cloning of the tool loop
(1,847 with-tools traces)

then

3

2 · GRPO on fresh rollouts
same loop, live tool execution
group-relative advantage

accuracy | format | tool use | efficiency

efficiency term exempts locate_tooth + nudge_crop

output:
LoRA adapters
loaded by the policy
at inference

rollouts execute the real dispatcher - same code path as trace generation
checkpoint selection on the dev split (test set evaluated once, at the end)

6

Evaluation (agent-loop figure)
dev-split signal returns to training

4

Policy + tool loop (agent-loop figure)
adapters loaded, findings evaluated

numbered path: 1 data -> traces · 2 corpus -> SFT · 3 SFT -> GRPO · 4 adapters -> policy ·
5 findings -> eval (agent-loop figure) · 6 dev-split signal